

WEB3 ECOSYSTEM USER BEHAVIOR & CHURN ANALYSIS

Project Summary Report

Prepared By:

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Tools Used:

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

Project Type:

Blockchain Data Analytics Project

1. PROJECT OVERVIEW

This project analysed blockchain wallet transaction behaviour using Python-based data analytics techniques. The analysis helps understand user engagement, wallet activity, transaction trends, and potential churn behaviour within a Web3 ecosystem.

The project demonstrates practical skills in:

- Data Cleaning
- Exploratory Data Analysis
- Data Visualization
- User Segmentation
- Behavioural Analytics

2. PROJECT OBJECTIVES

The main objectives of this project were:

- Analysed cryptocurrency transaction data
- Understand wallet activity patterns
- Identify active and inactive users
- Segment users using RFM analysis
- Detect potential churn behaviour
- Generate business insights from blockchain data

3. DATASET INFORMATION

The dataset contains blockchain transaction records including:

- Transaction Hash
- Sender Address
- Receiver Address
- Amount
- Gas Fee
- Timestamp
- Currency

- Transaction Status
- Gas Used
- Mining Pool

4. TECHNOLOGIES USED

Python was used for:

- Data preprocessing
- Data cleaning
- Exploratory Data Analysis
- User segmentation
- Visualization

Libraries Used:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

5. ANALYSIS PERFORMED

Data Cleaning:

- Missing value handling
- Duplicate removal

- Data type conversion
- Timestamp formatting

Exploratory Data Analysis:

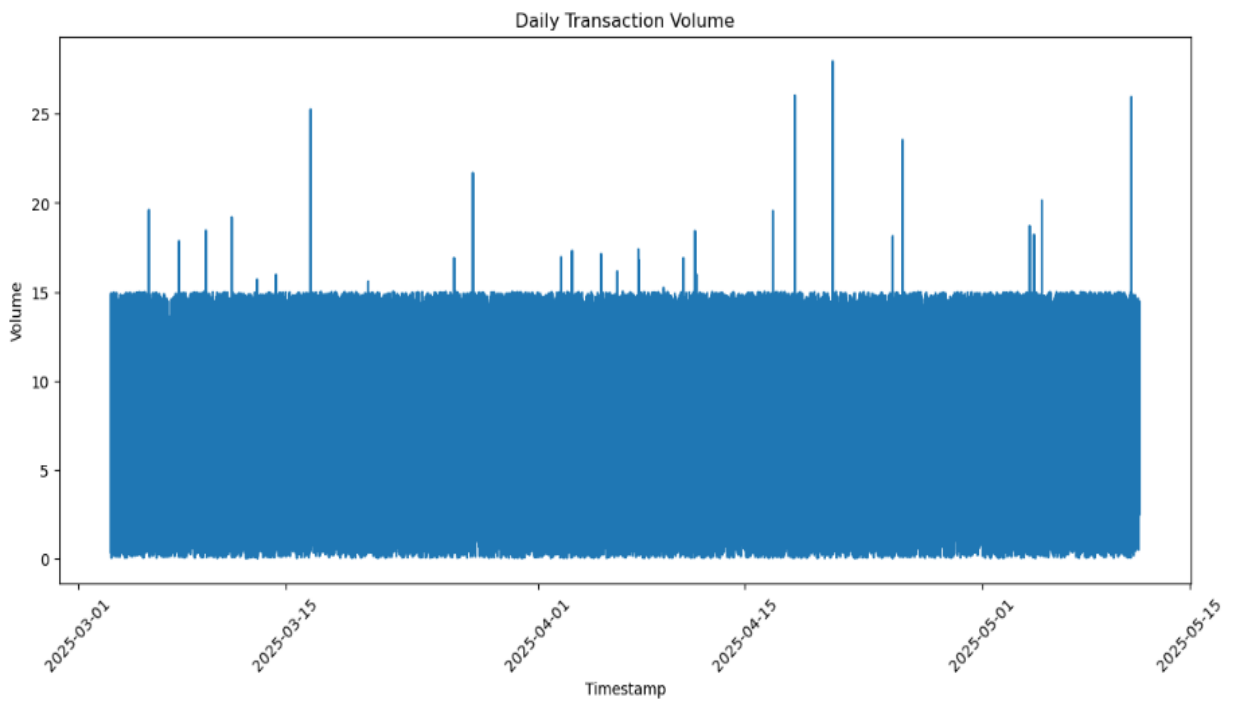
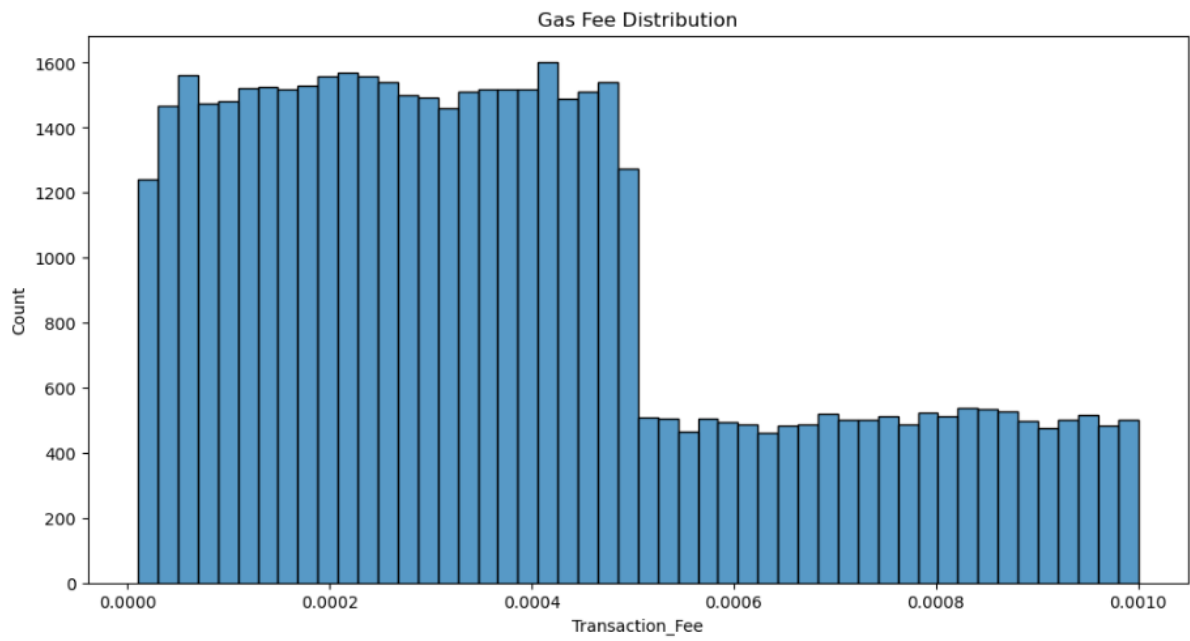
- Transaction volume trends
- Wallet activity distribution
- Gas fee patterns
- Currency usage
- Transaction success rates

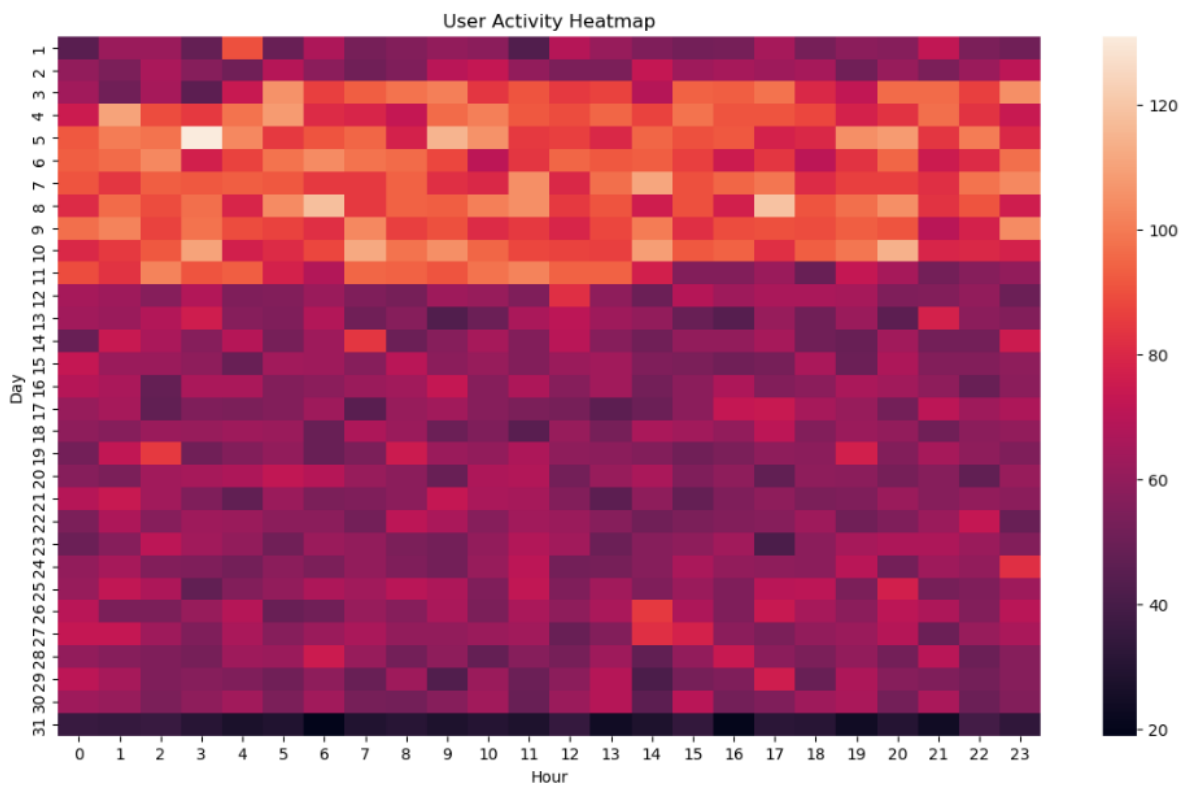
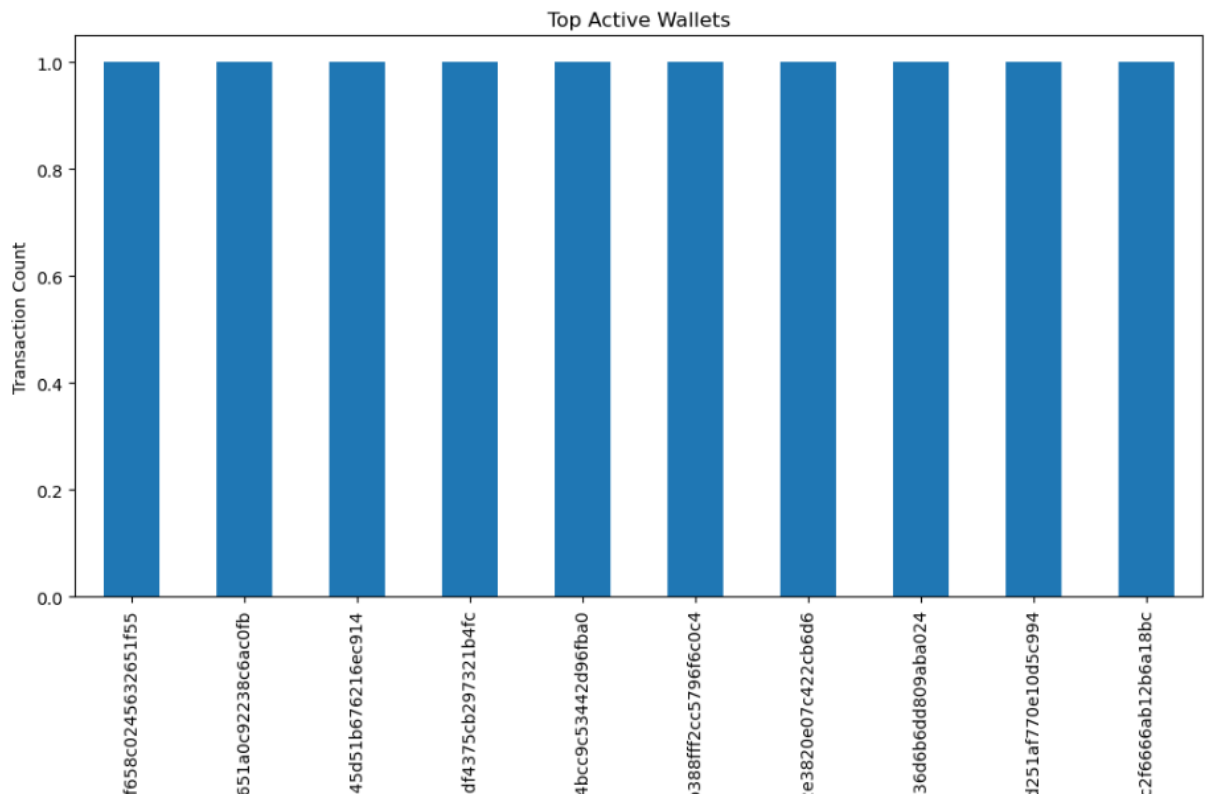
RFM Analysis:

- Recency Analysis
- Frequency Analysis
- Monetary Value Analysis

User Segmentation:

- Loyal Users
- High-Value Wallets
- Dormant Users
- Churn-Risk Users





6. KEY FINDINGS

- A small percentage of wallets contributed a large portion of transaction value.
- Many wallets showed irregular transaction behavior.
- Some users became inactive after periods of high activity.
- Higher gas fees reduced transaction frequency.
- RFM analysis identified high-value and churn-risk users.

7. BUSINESS INSIGHTS

Improve User Retention:

Platforms can target inactive wallets using:

- Rewards
- Incentives
- Reduced transaction fees

Monitor Churn Risk:

Wallets showing declining activity can be monitored for early churn detection.

Optimize Transaction Experience:

Reducing transaction costs may improve user engagement and ecosystem participation.

8. CONCLUSION

This project demonstrates how blockchain transaction data can be transformed into actionable insights using Python analytics and visualization techniques.

The analysis successfully identified:

- User behaviour patterns
- Wallet engagement trends
- High-value users
- Potential churn-risk wallets

The project highlights strong skills in:

- Data Analysis
- Data Visualization
- Behavioural Analytics
- Web3 Ecosystem Understanding

9. PROJECT DELIVERABLES

Included in the project:

- Jupyter Notebook
- Cleaned Dataset
- RFM Analysis Dataset
- Visual Analytics Charts

